

Bradley Nguyen

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science

Richardson, TX

May 2028

- **Relevant Coursework:** Computer Science II, Intro to Human Computer Interactions, Discrete Math, Systems Programming in UNIX, Object-Oriented Programming

EXPERIENCE

Parekh Lab for Motivational, Affective, and Cognitive Systems

Research Assistant

Richardson, TX

Sep 2025 – Present

- Shipped a full-stack experiment dashboard in **Next.js**, **TypeScript**, **FastAPI**, & **SQLAlchemy** that generated, compiled, & flashed Arduino firmware with **Jinja2** & **arduino-cli**, logged serial data with **pyserial**, and displayed live camera feeds in-browser for **6+ researchers** across **8 hardware rigs**
- Built a pose-estimation pipeline with **PyTorch** & **DeepLabCut** that raised frame-level detection from **35%** to **90%+** across **7 keypoints**, by fine-tuning **SuperAnimal** weights & optimizing detector thresholds and **CUDA** batch inference
- Developed a **SLEAP** tracking pipeline reaching **95%+ detection** across **26 multi-hour recordings**, debugging training failures across **4 rounds** to produce behavioral data the lab pairs with neural recordings to study motivated behavior

ACM UTD

Undergraduate Researcher

Richardson, TX

Jan 2026 – May 2026

- Engineered a defense pipeline hardening **LLM-based** vision-language models against **prompt injection**, cutting attack success from **90.9%** to **26%** on **Gemma-3** & **Qwen-2.5**, via input validation, prompt-level security boundaries, & prompt isolation in **Python** & **Hugging Face Transformers**
- Benchmarked **AI security** across **13 large vision-language models**, then evaluated **12 defenses** against **6 image-based injection attacks** on **Gemma-3**, **Llama-3.2**, & **Qwen-2.5** across **3,200+ medical scans** in **Python** with **GPU-accelerated inference**
- Sustained real-time inference at **0.315s** on **Qwen-2.5** during attack mitigation, by designing lightweight prompt-based filtering that blocked malicious visual overlays without modifying model architecture

Data Lake Group

Software Engineer Intern

Vienna, VA

Nov 2025 – Feb 2026

- Deployed **AWS** services including **S3**, **Lambda**, & **IAM** to power **15 automation workflows** that streamlined data processing across internal teams
- Automated repetitive provisioning tasks with **Python** scripts & **YAML** configs, lifting pipeline reliability **25%** & cutting manual setup by **8 hrs/wk**
- Authored documentation & shipped feature updates to internal cloud tooling, turning feedback from **5+ teammates** into a smoother provisioning workflow for the team

ACM UTD

Software Engineer

Richardson, TX

Aug 2025 – Dec 2025

- Delivered a Spotify-integrated full-stack music discovery & sharing platform with a team of **5**, presented to a panel of **5 industry judges**
- Optimized backend infrastructure, cutting database query latency **30%**, by building **RESTful APIs** with **Node.js**, **Express**, **TypeScript**, & **Supabase**
- Implemented cross-entity search & music-playback flows at **sub-200ms** response times, by integrating **Supabase** full-text queries with the **Spotify Web SDK**

PROJECTS

MEMEoTron (HackUTD 2025) | Python, Flask, Google Cloud Vision API, Nemotron

Nov 2025

- Extracted linguistic & spatial metadata from **1,000+ viral images** at **95% OCR accuracy** via a **Python** & **Google Cloud Vision** pipeline
- Cut media interpretation latency to **sub-300ms**, by building a **Flask RESTful API** that piped OCR data to web-scraping agents for the **Nvidia Nemotron** model

UTD Clubs - Campus Involvement Platform | NextAuth.js, Drizzle ORM, tRPC, Tailwind CSS

Oct 2025

- Secured access & data integrity for **500+ student users** with **NextAuth.js** authentication, **Drizzle ORM**, & type-safe **tRPC** APIs
- Redesigned mobile navigation, lifting usability scores **40%** across tablet & phone viewports, by refactoring **Radix UI** primitives with **Tailwind CSS**

TECHNICAL SKILLS

Languages: Java, C++, HTML, CSS, JavaScript, TypeScript, Python, SQL

Frameworks/Libraries: Next.js, React, Tailwind CSS, Node.js, Express.js, FastAPI, PyTorch, Hugging Face Transformers

Developer Tools: Git/GitHub, AWS, PostgreSQL, Postman, Supabase, Google Colab, Claude Code, Google Antigravity